

# Adaptive Fuzzy-PI/PID Tether Control for a UAV-UGV System

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**Abstract**— This paper presents the development of a model-based design approach for a tethered unmanned aerial vehicle (UAV) and unmanned ground vehicle (UGV) system. An integrated dynamic model is formulated to capture the coupled dynamics of the UAV, tether, and winch mechanism. A Sugeno-type fuzzy inference system (FIS) is employed to adaptively adjust the reference tether length based on wind velocity and the desired geometric separation between the UAV and UGV. A dual-loop PID/PI control strategy with switching logic is implemented to regulate both tether length and winch motor velocity, enabling robust tracking of the reference tether length. Simulation results demonstrate the effectiveness of the integrated.

**Keywords**—Unmanned aerial vehicles (UAVs), tethered UAV, tether, winch, PID, Autonomous Systems

## I. INTRODUCTION

Tethered Unmanned Aerial Vehicles (UAVs) have gained significant attention in recent years. Compared to free-flying UAVs, tethered systems overcome the limitations of onboard battery capacity and provide secure, interference-resistant communication links. These characteristics position tethered UAVs as a viable alternative for missions requiring extended operation and high communication reliability.

### A. Literature Review

Existing research highlights the suitability of tethered UAV platforms for long-duration missions such as surveillance, emergency response, and telecommunications, see [1], [2], [3] and [4]. The presence of a tether, however, significantly complicates the system dynamics. Its nonlinear and coupled behaviour, driven by catenary geometry and aerodynamic drag, introduces a dynamic constraint that strongly influences UAV motion, particularly under wind disturbances [5]. Studies by Nicotra et al. [5] and Yan et al. [6] model both taut and slack cable regimes, showing that cable mass and length variation induce oscillatory effects and additional control challenges. Marques et al. [3] further emphasised that effective tether management must address both mechanical stability and control adaptability in the presence of environmental perturbations.

Classical PID control has been the predominant approach for regulating UAV attitude and tether behaviour due to its simplicity and ease of implementation [7]. Yet fixed-gain PID controllers struggle with the nonlinearities and parameter variations introduced by changing tether forces and wind gusts. To overcome these limitations, Musa [8] reviewed quadcopter control strategies and showed that fuzzy logic and neural networks offer improved adaptability to real-time disturbances. Misin [9] demonstrated that Model Predictive Control (MPC) can maintain UAV stability by predicting future states and optimising control actions under constraints.

D'Souza et al. [10] extended this direction by modelling tethered UAV systems within advanced simulation environments, capturing aerodynamic coupling and enabling controller validation prior to real-world testing.

Cooperative UAV-UGV (unmanned ground vehicle) systems have also emerged as a promising direction, combining the aerial mobility of UAVs with the endurance and payload capacity of ground vehicles. Miki et al. [11] demonstrated that UAV assistance can enhance UGV performance in challenging terrain. When a tether is introduced into such architectures, new degrees of freedom and control dependencies arise. The UAV, tether, and winch must therefore be co-optimised to ensure stable and coordinated operation [12].

### B. Novelty of the Research

While the literature addresses elements of UAV attitude control and tether management, there is little work on the development of an integrated, model-based design that explicitly captures the interactions between the UAV, tether, and winch subsystems and couples this with an adaptive tether-length policy. This paper contributes:

- A unified dynamic model combining UAV rigid-body motion, tether behavior, and winch electromechanics.
- An adaptive tether-length strategy using a Sugeno-type fuzzy inference system (FIS) to maintain positive tension under wind.
- A coordinated dual-loop control architecture in which UAV stabilisation and winch operation are jointly managed.

### C. Outline of Paper

The paper is organised as follows. Section II presents the system overview and problem formulation. Section III details UAV dynamics and control. Section IV describes the overall tethered UAV modelling and control. The simulation results are presented in Section V. Conclusion and details of further work are given in Section VI.

## II. SYSTEM OVERVIEW AND PROBLEM FORMULATION

The tethered UAV-UGV system shown in Figure 1 consists of a UGV, a quadrotor UAV, and an interconnecting tether. The tether provides a continuous electrical power supply to the UAV and enables bidirectional data communication with the UGV. A motor-driven winch mounted on the UGV adjusts the tether length dynamically during operation via an actuated spool mechanism.

The system operates within a three-dimensional inertial frame. The UAV exhibits six degrees of freedom (DOF): translational motion along the inertial axes ( $X, Y, Z$ ) and

rotational motion about the body-fixed axes corresponding to roll ( $\phi$ ), pitch ( $\theta$ ), and yaw ( $\psi$ ). The UAV state vector at time  $t \in \mathbb{R}_{\geq 0}$  is defined as:

$$\chi(t) = [x(t), y(t), z(t), \phi(t), \theta(t), \psi(t)]^T \quad (1)$$

where  $\chi(t) \in \mathbb{R}^6$  comprises the translational position of the UAV's center of mass,  $\chi_{tra}(t) = [x(t), y(t), z(t)]^T$ , and the rotational coordinates represented by the Euler angles:  $\chi_{rot}(t) = [\phi(t), \theta(t), \psi(t)]^T$ .

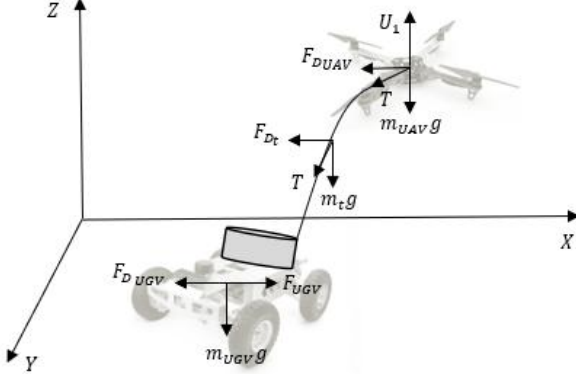


Fig 1: Free body diagram of tethered UAV-UGV system [5]

The UAV is actuated by four rotors, where each rotor  $i \in \{1, 2, 3, 4\}$  generates an upward thrust proportional to the square of its angular velocity  $\omega_i^2$ . The control input vector is defined as:  $U_{uav} = [U_1, U_2, U_3, U_4]^T$ , where the individual inputs are:

$$U_1 = b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \quad (2)$$

$$U_2 = bl \sin\left(\frac{\pi}{4}\right) \cdot (\omega_1^2 + \omega_4^2 - \omega_3^2 - \omega_2^2) \quad (3)$$

$$U_3 = bl \sin\left(\frac{\pi}{4}\right) \cdot (-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2) \quad (4)$$

$$U_4 = d(-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2) \quad (5)$$

where  $U_1$  denotes the total thrust,  $U_2$  and  $U_3$  denote the roll and pitch torques, respectively, and  $U_4$  denotes the yaw torque. The parameters,  $b$  and  $d$  denote the rotor thrust and drag coefficients, and  $l$  denotes the distance from the UAV centre to each rotor. The instantaneous tether length  $L$  denotes the computed as the Euclidean distance between the UAV and UGV positions:

$$L = \sqrt{(x - x_{UGV})^2 + (y - y_{UGV})^2 + (z - z_{UGV})^2} \quad (6)$$

The tethered UAV-UGV system is influenced by gravity, wind-induced aerodynamic drag, actuator limits, and bounds on allowable tether length. The tether imposes a geometric constraint on the UAV's position relative to the UGV, and these constraints must be respected during operation. The goal is to design a control strategy that:

- enables the UAV to track a desired trajectory,
- regulates the tether length through winch actuation, and
- maintains stability and robustness under wind disturbances.

To support the modelling and controller design, the following assumptions are adopted:

1. The airframe and components are symmetric.
2. Environmental disturbances other than wind (e.g., thermal effects) are neglected.
3. The UAV mass is constant and uniformly distributed.
4. The UAV behaves as a rigid body without deformation.
5. The centre of gravity is located at the body-frame origin.
6. Thrust and drag forces are proportional to the square of the rotor speeds.

### III. UAV DYNAMICS AND CONTROL

#### A. UAV Dynamics

The UAV orientation is defined through the transformation between the body-fixed frame and the inertial frame. This relationship is represented by the rotational transformation matrix  $R \in \mathbb{R}^{3 \times 3}$  [9], which is used to project the thrust vector into the inertial frame. The matrix is given by:

$$R = \begin{bmatrix} c\psi c\theta & s\phi s\theta c\psi - c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\phi c\psi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{bmatrix} \quad (7)$$

where  $c$  is the cosine and  $s$  is the sine of the Euler angle. The translational motion of the UAV is described by the Newton-Euler equations:

$$m\ddot{\chi}_{tra} = RF_z + F_g + F_{D UAV} \quad (8)$$

where  $\ddot{\chi}_{tra} \in \mathbb{R}^3 = [\ddot{x}, \ddot{y}, \ddot{z}]^T$  is translational acceleration,  $F_z \in \mathbb{R}^3$  is the thrust vector generated by the rotors,  $F_g \in \mathbb{R}^3$  is the gravitational force, and  $F_{D UAV} = \frac{1}{2} \rho_a A_x C_d (\dot{\chi}_{tra} + V_w)^2$  is the aerodynamic drag, where  $V_w = [V_{wx}, V_{wy}, V_{wz}]^T$  denote the wind velocity components. Using (8), the translational dynamics become:

$$\ddot{x} = \frac{(s\psi s\phi + c\psi s\theta c\phi)U_1 - F_{D UAV x}}{m} \quad (9)$$

$$\ddot{y} = \frac{(-c\psi s\phi + s\psi s\theta c\phi)U_1 - F_{D UAV y}}{m} \quad (10)$$

$$\ddot{z} = -g - \frac{(c\theta c\phi)U_1 - F_{D UAV z}}{m} \quad (11)$$

The rotational dynamics follow Euler's rigid-body equations:

$$I \dot{\omega} + \omega(I\omega) = \tau \quad (12)$$

where  $\omega \in \mathbb{R}^3$  is the angular velocity in the body frame,  $I = \text{diag}(I_{xx}, I_{yy}, I_{zz}) \in \mathbb{R}^{3 \times 3}$  is the inertia matrix, and  $\tau = [\tau_x, \tau_y, \tau_z]^T = [U_2, U_3, U_4] \in \mathbb{R}^3$  is the applied torque vector. Expanding (12) yields:

$$\ddot{\phi} = \frac{I_{xx}(I_{yy} - I_{zz})\dot{\theta}\dot{\psi} - J_r\dot{\theta}\omega_r + U_2}{I_{xx}} \quad (13)$$

$$\ddot{\theta} = \frac{(I_{zz} - I_{xx})\dot{\psi}\dot{\phi} + J_r\dot{\phi}\omega_r + U_3}{I_{yy}} \quad (14)$$

$$\ddot{\psi} = \frac{(I_{xx} - I_{yy})\dot{\phi}\dot{\theta} + U_4}{I_{zz}} \quad (15)$$

where  $J_r$  denotes the rotor inertia and  $\omega_r$  denotes the resultant rotor speed.

### B. Trajectory Tracking Control of the UAV

A cascaded control architecture, see Figure 2 is used to track a desired trajectory. The outer loop controls UAV position, and the inner loop stabilises orientation.

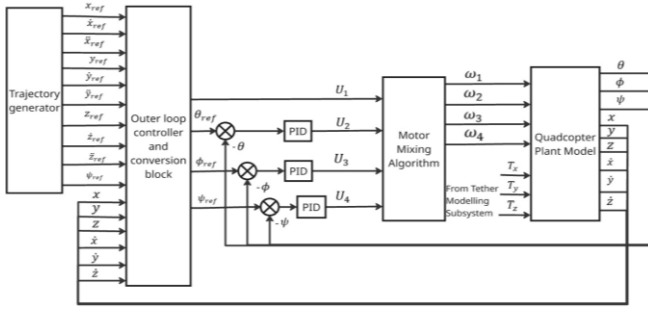


Fig 2: UAV system Architecture.

#### 1) Position Controller

The desired position and velocity vectors are  $\chi_{tra_d} = [x_d, y_d, z_d]^T \in \mathbb{R}^3$  and  $\dot{\chi}_{tra_d} = [\dot{x}_d, \dot{y}_d, \dot{z}_d]^T \in \mathbb{R}^3$ , respectively. Position and velocity errors defined as  $e_{\chi_{tra}} = \chi_{tra_d} - \chi_{tra}$  and  $\dot{e}_{\chi_{tra}} = \dot{\chi}_{tra_d} - \dot{\chi}_{tra}$ , where  $e_{\chi_{tra}} = [e_x \ e_y \ e_z]^T \in \mathbb{R}^3$  and  $\dot{e}_{\chi_{tra}} = [\dot{e}_x \ \dot{e}_y \ \dot{e}_z]^T \in \mathbb{R}^{3 \times 1}$ . Here,  $e_x, e_y$ , and  $e_z$  are the position errors of the UAV, and  $\dot{e}_x, \dot{e}_y$ , and  $\dot{e}_z$  are the velocity errors of the UAV. To track the desired trajectory, a second-order state-feedback position controller is implemented [10]. The control objective is to compute an inertial acceleration command  $U_{outerloop} = K[e_p^T, \dot{e}_p^T]^T = [U_x, U_y, U_z]^T \in \mathbb{R}^3$ , with this given by:

$$U_{outerloop} = \begin{bmatrix} K_{x1} & K_{x2} \\ K_{y1} & K_{y2} \\ K_{z1} & K_{z2} \end{bmatrix} \begin{bmatrix} e_x & \dot{e}_x \\ e_y & \dot{e}_y \\ e_z & \dot{e}_z \end{bmatrix} \quad (16)$$

where  $K \in \mathbb{R}^{3 \times 2}$  denotes the state feedback gain matrix, consisting of the individual gains  $K_{x1}, K_{y1}, K_{z1}, K_{x2}, K_{y2}$ , and  $K_{z2}$ . To formulate this in terms of second-order error dynamics, the control input  $U_{outerloop}$  is equated to the second derivative of the position error. These second order different equations are thus simplified into its time-variant form, with these given by:

$$e(t) = C\tilde{e}^{\lambda t} \quad (17)$$

$$e(t) = A\tilde{e}^{\lambda_1 t} + B\tilde{e}^{\lambda_2 t} \quad (18)$$

where  $\lambda_1$  and  $\lambda_2$  are the desired pole locations. Hence, the gains are computed as follows:

$$K_2 = \lambda_1 + \lambda_2 \quad (19)$$

$$K_1 = \left( \lambda_1 - \frac{\lambda_1 + \lambda_2}{2} \right)^2 - \frac{(\lambda_1 + \lambda_2)^2}{4} \quad (20)$$

To convert the desired accelerations into attitude references, the standard mapping is used:

$$\theta_d = \tan^{-1} \left[ \frac{\ddot{z}}{\ddot{z} + g} c\psi_d + \frac{\dot{y}}{\ddot{z} + g} s\psi_d \right] \quad (21)$$

$$\phi_d = \tan^{-1} \left[ \frac{c\theta \cdot \frac{\ddot{x}}{\ddot{z} + g} - t\theta c\psi_d}{N\psi_d} \right] \quad (22)$$

where:

$$\psi_d = \tan^{-1}(\dot{y}_d/\dot{x}_d)$$

is the desired yaw angle, wrapped between  $[-\pi, +\pi]$ . In (22), the term  $N\psi_d$  denote the trigonometric component by  $\psi_d$ :

- If  $\frac{3\pi}{4} \leq \psi_d \leq \frac{\pi}{4}$ , the denominator  $N\psi_d$  uses  $c\psi_d$
- Otherwise,  $N\psi_d$  uses  $s\psi_d$

#### 2) Inner Loop

The inner-loop controller stabilises the UAV's orientation by ensuring the actual Euler angles track the desired references ( $\chi_{rot_d} = [\phi_d, \theta_d, \psi_d]^T$ ). Let orientation error  $e_{\chi_{rot}} = \chi_{rot_d} - \chi_{rot} \in \mathbb{R}^3$ . In this case, as displayed in Figure 2, a PID control structure is implemented for each rotational axis. For each axis, the corresponding control input  $U_{rot,i} \in \{U_2, U_3, U_4\}$  is given by:

$$U_{rot,i} = K_{p_{\chi_{rot}}} (e_{\chi_{rot}}) + K_{d_{\chi_{rot}}} (\dot{e}_{\chi_{rot}}) + K_{i_{\chi_{rot}}} \int (e_{\chi_{rot}}) dt \quad (23)$$

where  $i = 2$  for  $\chi_{rot} = \phi$ ,  $i = 3$  for  $\chi_{rot} = \theta$ , and  $i = 4$  for  $\chi_{rot} = \psi$ . Furthermore,  $K_{p_{\chi_{rot}}}$ ,  $K_{d_{\chi_{rot}}}$ , and  $K_{i_{\chi_{rot}}}$  are the proportional, derivative and integral control gains respectively.

## IV. MODELLING OF TETHER CONTROL SYSTEM

This section develops the modelling of the tether control subsystem within the UAV-UGV architecture illustrated in Figure 3. The modelling includes the tether dynamics and the winch actuator, while the control strategy integrates a dual-loop controller, a FIS, and a switching mechanism to generate appropriate tether length references.

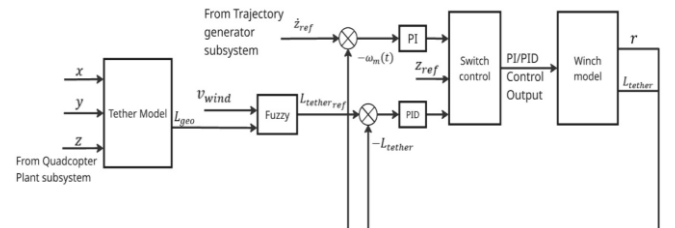


Fig 3: Tether system architecture.

### A. Tether Modelling and Control

Although the tether is often approximated as inextensible, in this work it is modelled as a linear elastic spring subject to axial deformation. This approximation accounts for the additional tether extension needed under wind disturbances to prevent slack. Following Hooke's Law, and neglecting bending stiffness, damping, and temperature-dependent effects as in [11], the tether tension,  $T_0$ , is given by:

$$T_0 = K_t(L - L_0) \quad (24)$$

where  $K_t$  is the tether stiffness coefficient,  $L$  is the instantaneous tether length and  $L_0$  is the unstretched tether length. To incorporate the additional extension required under aerodynamic loading, the extra length, denoted  $\Delta L$ , is computed from the drag forces acting on both the UAV and the tether. The resulting formulation is:

$$T_D = F_{D_{UAV}} + F_{D_t} \quad (25)$$

$$\Delta L = \frac{F_{D_{UAV}} + F_{D_t}}{K_t} \quad (26)$$

where  $T_D$  is the tension induced by combined aerodynamic drag on the UAV and the tether.

### B. Winch Modelling

The winch subsystem is represented by using coupled electrical and mechanical dynamics [11], capturing motor inductance, resistance, back-EMF, friction, pulley radius, and inertia. The electrical dynamics of the DC motor are described by:

$$V(t) = L_a \cdot \frac{di(t)}{dt} + R_a i(t) + K_e \omega_m(t) \quad (27)$$

where  $V(t)$  denotes the applied motor voltage,  $i(t)$  denotes the armature current,  $L_a$  denotes the armature inductance,  $R_a$  denotes the armature resistance,  $K_e$  denotes the back-EMF constant, and  $\omega_m(t)$  denotes the motor angular velocity. The mechanical subsystem of the model is given as:

$$k_t \cdot i(t) = J \cdot \frac{d\omega_m(t)}{dt} + B_f \omega_m(t) + T_0 \quad (28)$$

where  $k_t$  denotes the torque constant,  $J$  denotes the spool moment of inertia,  $B_f$  denotes the friction coefficient and  $T_0$  denotes the tether tension from (24). Once the physical prototype is constructed, with this viewed as future work, system identification techniques will be employed to refine these parameters and enhance the accuracy of the winch model, thereby improving the overall tether control performance.

### C. Fuzzy Inference System for Tether Length Control

A Sugeno-Type 1 FIS is implemented to adaptively compute the desired tether length under varying operational conditions. The FIS uses two input variables:

The input variables are represented using evenly spaced triangular membership functions, as shown in Figure 4. UAV altitude is partitioned into three fuzzy sets:

- *Short (S)*,
- *Medium (M)*
- *Long (L)*

The wind velocity partitioned into:

- *Calm (C)*
- *Moderate (MO)*
- *High (H)*

These sets capture typical variations in altitude and wind loading encountered during operation. The output variable uses two Sugeno-type consequents:

- Base (B): a constant output
- Wind Adjusted (W): a linear function accounting for additional tether extension under aerodynamic loading

Given three membership functions for each input, the rule base consists of  $3 \times 3 = 9$  rules derived from their Cartesian product. Rules are activated via the AND (minimum) operator. Example rules include:

*If  $L_{geo}$  is short (S) and  $V_{wind}$  is calm (C), then  $L_{ref}$  is Base (B).*  
*If  $L_{geo}$  is long (L) and  $V_{wind}$  is high (H), then  $L_{ref}$  is Wind Adjusted (W)*

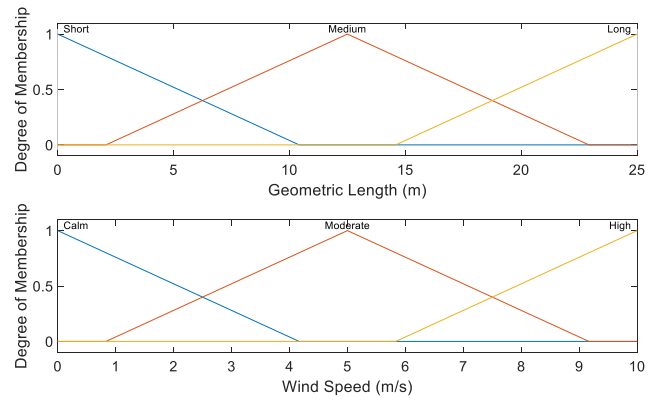


Fig 4: Input membership function.

The fired rule consequents are aggregated using the maximum operator, and the final scalar output is computed using the weighted average defuzzification method:

$$L_{ref} = \frac{\sum_{i=1}^n w_i \cdot z_i}{\sum_{i=1}^n w_i} \quad (29)$$

where  $w_i$  denotes the firing strength of rule  $i$ ,  $z_i$  denotes the singleton output. The resulting FIS output surface is shown in Figure 5.

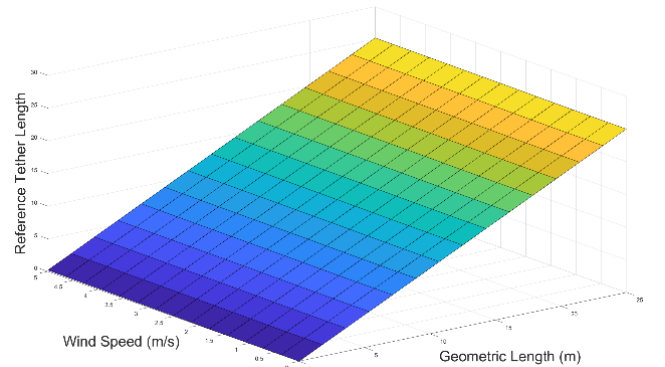


Fig 5: Fuzzy logic control surface

#### D. Winch Control

A dual-loop control strategy is employed to ensure safe and efficient tether management. The objective is to synchronise tether deployment with UAV motion while allowing rapid actuation when large altitude changes occur.

##### 1) Altitude Control Loop

A PID controller regulates the tether-length error,  $e_{loop_1} = L_{ref} - L$ , where  $L = r_c \theta_c$ ,  $r_c$  is the spool radius, and  $\theta_c$  is the angular position of the winch motor. During steady operation this loop provides precise tracking of the FIS-generated reference length  $L_{ref}$ .

##### 2) Spool Speed Control Loop

A PI controller regulates the motor angular velocity  $\omega_m(t)$  based on the error,  $e_{loop_2} = \omega_m(t) - \dot{z}/r_c$ . This loop synchronises tether deployment or retraction with the UAV's vertical motion, ensuring smooth operation during ascent or descent.

##### 3) Switching Logic

A logic mechanism selects the appropriate control mode. When the difference between the commanded final altitude  $r_{au}$  and the UAV's current altitude  $z$  exceeds a predefined threshold, the PI speed controller is activated to achieve rapid tether deployment. Once the error falls below the threshold, control switches back to the PID length controller to ensure accurate steady-state tracking.

#### E. Tethered UAV Dynamics

The coupled dynamics of the tethered UAV are influenced by gravity, aerodynamic drag, and the tension components ( $T_x, T_y, T_z$ ) transmitted through the tether. The resulting translational dynamics are:

$$\ddot{x} = \frac{(s\psi s\phi + c\psi s\theta c\phi)U_1 - F_{D_{UAV_x}} - T_x}{m} \quad (30)$$

$$\ddot{y} = \frac{(-c\psi s\phi + s\psi s\theta c\phi)U_1 - F_{D_{UAV_y}} - T_y}{m} \quad (31)$$

$$\ddot{z} = -g - \frac{(c\theta c\phi)U_1 - F_{D_{UAV_z}} - T_z}{m} \quad (32)$$

## V. RESULTS

A simulation study was conducted in MATLAB and Simulink to evaluate the performance of the proposed tether control strategy for the UAV-UGV system. The test scenario consisted of an ascent to a height of 3m followed by circular motion with a radius of 0.5m. The scenario was tested under three wind conditions: 0m/s, 5m/s, and 10m/s. These cases provide a rigorous basis for assessing UAV trajectory tracking and tether management performance.

The Integral of Absolute Error (IAE) was used as the primary performance metric:

$$IAE = \int_0^T |e(t)|dt \quad (33)$$

where  $e(t)$  is the tracking error between the desired and actual response, and  $T$  is the total simulation time. The UAV, tether, and winch parameters used for the simulations are

presented in Table 1, derived from [5] and [11]. The controller gains for the PID and PI loops are listed in Table 2.

TABLE 1: MODELLING PARAMETERS

| Modelling parameter | Value [units]                             |
|---------------------|-------------------------------------------|
| $\mu$               | 0.0022 [kg/m]                             |
| $m$                 | 1.15 [kg]                                 |
| $D_t$               | $3.5 \times 10^{-3}$ [m]                  |
| $C_{D_t}$           | 0.7                                       |
| $d_{CG}$            | 0.1 [m]                                   |
| $\rho_{air}$        | 1.225 [kg/m <sup>3</sup> ]                |
| $A_x$               | 0.1209 [m <sup>2</sup> ]                  |
| $A_y$               | 0.1209 [m <sup>2</sup> ]                  |
| $A_z$               | 0.1624 [m <sup>2</sup> ]                  |
| $C_D$               | diag(0.1 0.1 0.15)                        |
| $g$                 | 9.81 [m/s <sup>2</sup> ]                  |
| $I_x$               | $4.6 \times 10^{-3}$ [kg/m <sup>2</sup> ] |
| $I_y$               | $4.6 \times 10^{-3}$ [kg/m <sup>2</sup> ] |
| $I_z$               | $9.1 \times 10^{-3}$ [kg/m <sup>2</sup> ] |
| $l$                 | 0.225 [m]                                 |
| $k_t$               | 35 [N/m]                                  |
| $R$                 | 0.062 [ohms]                              |
| $L$                 | 0.02 [H]                                  |
| $rc$                | 0.05 [m]                                  |
| $B$                 | $6.7 \times 10^{-4}$ [Nms]                |
| $J$                 | 0.1 [Kg m <sup>2</sup> ]                  |

TABLE 2: PID/PI CONTROLLER GAINS

| Controlled Variable | Controller Gain |         |         |
|---------------------|-----------------|---------|---------|
|                     | $K_p$           | $K_i$   | $K_d$   |
| $x$                 | 0.0100          | 0.0001  | 0.0300  |
| $y$                 | 0.0100          | 0.0001  | 0.0300  |
| $z$                 | 0.0037          | 0.0003  | 0.0127  |
| $\omega_m(t)$       | 4.1489          | 22.2904 | N/A     |
| $L$                 | 80.000          | 30.0000 | 20.0000 |

Following the simulation setup, the UAV path-tracking performance and the corresponding tether control responses are shown in Figures 6 and 7. Table 3 summarises the tracking accuracy using the IAE metric. As expected, increasing wind velocity leads to higher IAE values in the  $X$  and  $Y$  directions due to greater lateral aerodynamic disturbance. The  $Z$ -axis IAE remains unchanged across all wind conditions, indicating that the altitude controller maintains consistent performance. At 10m/s, a pronounced increase in the  $Y$ -direction IAE is observed, reflecting the dominant influence of crosswind loading on lateral motion.

TABLE 3: POSITION (X,Y,Z) IAE

| Wind Velocity in the y-axis [m/s] | IAE  |       |      |
|-----------------------------------|------|-------|------|
|                                   | x    | y     | z    |
| 0                                 | 1.43 | 0.94  | 0.02 |
| 5                                 | 1.60 | 3.74  | 0.02 |
| 10                                | 3.03 | 13.68 | 0.02 |

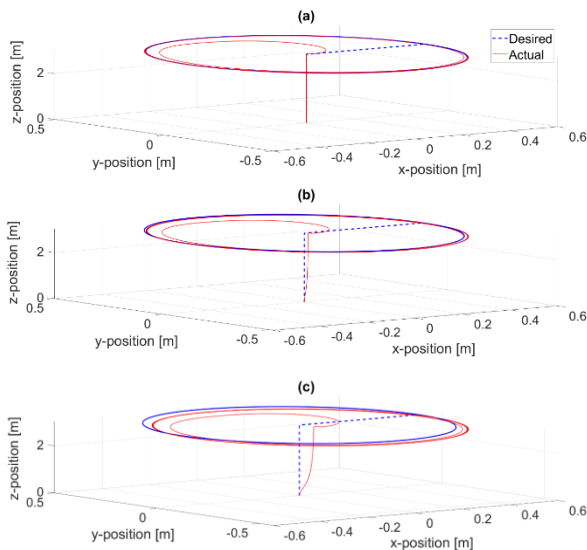


Figure 6: Effect of wind (a) 0 m/s. (b) 5 m/s (c) 10 m/s on trajectory

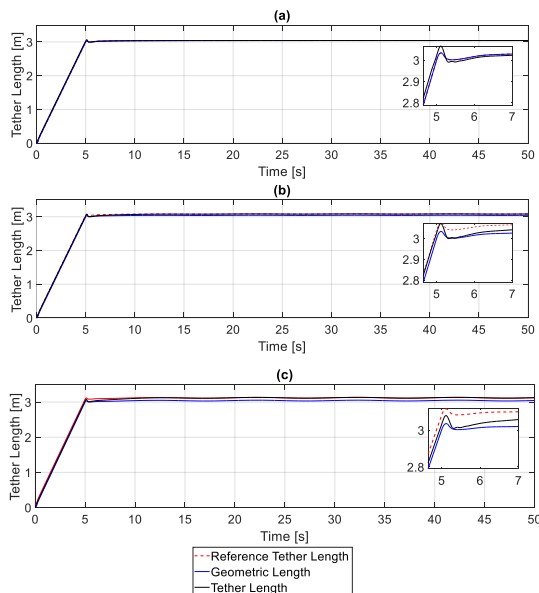


Fig 7: Effect of wind (a) 0 m/s. (b) 5 m/s (c) 10 m/s on tether length

The spiral trajectories for the three wind conditions, shown in Figure 6, provide a clear visualisation of the UAV's motion. In the zero-wind case (Figure 6(a)), the UAV tracks the desired Z-axis ascent and maintains accurate X – Y spiral motion throughout the test. When the wind speed increases to 5 m/s and 10 m/s (Figures 6(b) and 6(c)), the UAV is displaced in the Y direction, deviating from the planned path due to the lateral wind loading.

Tether control performance under the same conditions is presented in Figure 7. As wind speed increases, the fuzzy logic controller generates a reference tether length that exceeds the geometric length, compensating for aerodynamic drag and ensuring the tether remains under positive tension. At 0 m/s, the reference and geometric lengths coincide (Figure 7(a)), while at 5 m/s and 10 m/s (Figures 7(b) and 7(c)) the reference length increases appropriately to prevent slack.

The behaviour of the dual-loop winch controller is also evident in the results. During the initial 4 seconds, the large altitude error activates the PI speed loop, producing a smooth and rapid deployment response. Once the error falls below the switching threshold, control transitions to the PID length loop, yielding faster convergence to the reference value. At higher wind speeds, the increased reference extension commanded by the fuzzy logic leads to a modest overshoot during the PID phase, although the system maintains stable tracking with negligible steady-state error.

## VI. CONCLUSION

An integrated, model-based design for a tethered UAV–UGV system was presented, combining UAV dynamics, tether elasticity, and winch electromechanics with a coordinated control architecture. A Sugeno-type FIS adaptively commanded tether length to maintain positive tension under wind, while a dual-loop PID/PI strategy with switching logic balanced rapid deployment with precise length regulation. Simulations showed robust tracking and effective tether management across wind conditions. Future work will extend the outer loop with MPC for disturbance anticipation and constraint handling and will refine parameters using system identification on a physical prototype prior to real-world testing.

## REFERENCES

- [1] Rodriguez, V., Tipantuña, C., Reinoso, D., Carvajal-Rodriguez, J., Egas Acosta, C., Proaño, P. and Hesselbach, X., 2025. An Experimental Tethered UAV-Based Communication System with Continuous Power Supply. *Future Internet*, 17(7), p.273.
- [2] Fattori, F. and Cocuzza, S., 2025. Tethered Drones: A Comprehensive Review of Technologies, Challenges, and Applications. *Drones*, 9(6), p.425.
- [3] Marques, M.N., Magalhães, S.A., Dos Santos, F.N. and Mendonça, H.S., 2023. Tethered unmanned aerial vehicles—a systematic review. *Robotics*, 12(4), p.117.
- [4] Madrigal, A.C., 2011. Inside the drone missions to Fukushima. *The Atlantic*, 28 April
- [5] Nicotra, M.M., Naldi, R. and Garone, E., 2017. Nonlinear control of a tethered UAV: The taut cable case. *Automatica*, 78, pp.174-184.
- [6] Yan, C., Yang, J., Zhang, D., Zhang, S. and Zhang, T., 2025. Comprehensive Design and Experimental Validation of Tethered Fixed-Wing Unmanned Aerial Vehicles. *Aerospace* (MDPI Publishing), 12(2).
- [7] Abdelhay, S. and Zakriti, A., 2019. Modeling of a quadcopter trajectory tracking system using PID controller. *Procedia Manufacturing*, 32, pp.564-571.
- [8] Musa, S., 2018. Techniques for quadcopter modeling and design: A review. *Journal of unmanned system Technology*, 5(3), pp.66-75.
- [9] Misin, M., 2020. Model predictive controller of a UAV using the LPV approach (Master's thesis, Universitat Politècnica de Catalunya).
- [10] D'Souza, J., Burnham, K., Kavakli-Thorne, M. and Pickering, J.E., 2024, November. Modelling and Simulation of an Autonomous Pod-Tethered Quadcopter Drone System for Aviation Applications. In *Proceedings of the 21st International Conference on Informatics in Control, Automation and Robotics (ICINCO 2024)*, Porto, Portugal (pp. 18-20).
- [11] Miki, T., Khrapchenkov, P. and Hori, K., 2019. UAV/UGV Autonomous Cooperation: UAV assists UGV to climb a cliff by attaching a tether. *arXiv preprint arXiv:1903.04898*.
- [12] Pickering, J.E., Ghanami, N., D'Souza, J., Kattyayam, S., Kavakli-Thorne, M. and Burnham, K.J., 2024, July. System Requirements and Review for the Operation of a Tethered Drone with an Autonomous Vehicle. In *2024 10th International Conference on Control, Decision and Information Technologies (CoDIT)* (pp. 2259-2265). IEEE.